

Funding & Innovation Platform

1. Title: Research Funding & Innovation Intelligence Platform

2. Objective

Build an AI-powered Research Funding & Innovation Intelligence Platform that helps researchers, startups, universities, innovation centers, and enterprises identify relevant funding opportunities, analyze research trends, evaluate patent landscapes, discover emerging technologies, and generate commercialization recommendations.

The platform combines research intelligence, funding opportunity discovery, patent analytics, technology trend analysis, and innovation strategy recommendations through a centralized innovation intelligence dashboard.

Outcomes

- Developed and deployed an AI-powered innovation intelligence platform.
 - Implemented secure authentication and role-based access control.
 - Built funding opportunity discovery and recommendation workflows.
 - Developed research trend analysis and technology intelligence modules.
 - Implemented patent landscape analysis and intellectual property analytics.
 - Built innovation scoring and commercialization recommendation systems.
 - Developed dashboards for funding, research, patent, and innovation insights.
 - Deployed the platform using Docker and cloud deployment platforms such as AWS or Azure.
-

3. Architecture Diagram:



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- User registration and login
- JWT authentication
- OAuth2 login
- Role-based access control
- User profile management

Roles

- Researcher
- Startup Founder
- Innovation Manager

- Administrator

2. Research Profile Management

- Research profile creation
- Research interest management
- Publication management
- Academic profile tracking
- Research history management

Profile Information

- Research Domains
- Keywords
- Publications
- Patents
- Technology Areas
- Organization Information

3. Funding Opportunity Discovery Module

- Funding opportunity collection
- Funding recommendation engine
- Eligibility matching
- Grant search
- Funding alerts

Funding Sources

- Government Grants
- Research Councils
- Innovation Funds

- Startup Accelerators
- Venture Programs
- International Funding Agencies

4. Research Trend Intelligence Module

- Publication trend analysis
- Emerging topic detection
- Research hotspot identification
- Domain trend monitoring
- Citation analytics

Data Sources

- Research Papers
- Publications
- Conference Proceedings
- Open Research Repositories

5. Patent Landscape Analysis Module

- Patent search
- Patent clustering
- Patent trend analysis
- Competitor patent analysis
- Innovation mapping

Patent Information

- Patent Title
- Assignee
- Filing Date

- Patent Classification
- Technology Domain
- Citation Count

6. Technology Intelligence Module

- Emerging technology identification
- Technology maturity analysis
- Technology adoption tracking
- Innovation opportunity discovery
- Competitive technology monitoring

7. Innovation Scoring Engine

- Innovation potential scoring
- Research impact scoring
- Technology readiness scoring
- Commercial viability scoring
- Funding attractiveness scoring

Weighted Scoring Model

Innovation Score =

- Research Novelty (30%)
- Patent Strength (20%)
- Technology Maturity (15%)
- Market Potential (20%)
- Funding Relevance (15%)

8. Commercialization Recommendation Module

- Research commercialization analysis

- Productization recommendations
- Licensing opportunities
- Startup creation recommendations
- Industry partnership suggestions

9. Dashboard & Analytics

Researcher Dashboard

- Funding recommendations
- Research trends
- Publication analytics
- Patent insights
- Innovation score

Startup Dashboard

- Funding opportunities
- Technology opportunities
- Patent intelligence
- Commercialization insights

Innovation Manager Dashboard

- Portfolio analytics
- Innovation pipeline tracking
- Technology trend monitoring
- Funding analytics

Admin Dashboard

- User management
- Platform analytics

- Recommendation monitoring
- System reports

10. Notification & Alert System

- New funding alerts
- Patent monitoring alerts
- Emerging technology alerts
- Research trend updates
- Commercialization opportunities
- Platform notifications

11. Reports & Export System

- Funding reports
- Patent reports
- Research trend reports
- Innovation intelligence reports
- Commercialization reports
- PDF export
- Excel export

12. Final Integration, Testing & Deployment

- Frontend and backend integration
- API validation and testing
- End-to-end workflow testing
- Security testing
- Performance optimization
- Docker containerization
- Production deployment

- Monitoring and logging setup
 - Documentation and user guides
-

5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

Tasks

- Define project objectives and innovation intelligence workflows.
- Design system architecture and database schema.
- Create UI wireframes and workflow planning.
- Setup frontend and backend environments.
- Implement authentication and role-based access system.
- Build research profile management workflows.
- Integrate publication and patent datasets.

Outcomes

- Understand innovation intelligence and funding discovery workflows.
- Learn system architecture and database design concepts.
- Build frontend and backend project initialization.
- Working authentication and profile management system.

Milestone 2: Week 3 & 4 — Funding Discovery & Research Intelligence

Tasks

- Implement funding recommendation engine.
- Build grant matching workflows.

- Develop publication trend analysis.
- Create research intelligence dashboards.
- Generate funding opportunity recommendations.

Outcomes

- Funding recommendation system operational.
- Research intelligence workflows functional.
- Trend analysis modules completed.

Milestone 3: Week 5 & 6 — Patent Analytics & Innovation Intelligence

Tasks

- Implement patent landscape analysis.
- Build technology intelligence engine.
- Develop innovation scoring workflows.
- Generate commercialization recommendations.
- Create innovation analytics dashboards.

Outcomes

- Patent intelligence operational.
- Innovation scoring system functional.
- Commercialization recommendation workflows completed.

Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

Tasks

- Build executive dashboards.
- Add reports and visualization modules.
- Implement testing and validations.

- Deploy platform using Docker and cloud services.
- Prepare final documentation and presentation.

Outcomes

- Fully deployed production-ready platform.
 - Innovation intelligence systems operational.
 - Complete end-to-end funding and innovation workflow demonstrable.
-

6. Evaluation Criteria

Milestone 1 (Week 2)

- Project initialization completed.
- Authentication implemented.
- Research profile management functional.
- Publication and patent datasets integrated.

Milestone 2 (Week 4)

- Funding recommendation engine operational.
- Research intelligence dashboards functional.
- Trend analysis workflows implemented.

Milestone 3 (Week 6)

- Patent intelligence operational.
- Innovation scoring functional.
- Commercialization recommendation workflows completed.

Milestone 4 (Week 8)

- Fully deployed frontend and backend.
- Dashboards and reporting systems operational.

- End-to-end innovation intelligence workflow demonstrated.
-

7. Tools & Tech Stack

Programming Language

Backend

- Python
- FastAPI

Frontend

- JavaScript
- React.js

Database

Primary Database

- PostgreSQL

Secondary Database

- MongoDB

AI & Machine Learning

- Scikit-learn
- XGBoost
- TensorFlow
- PyTorch
- Pandas
- NumPy

NLP & Research Intelligence

- Sentence Transformers
- Hugging Face Transformers
- LangChain
- OpenAI APIs

Search & Analytics

- Elasticsearch
- FAISS
- Vector Database

Research & Patent Data Sources

- OpenAlex
- CrossRef
- Semantic Scholar
- Google Patents
- The Lens
- USPTO Public Data

Cloud & DevOps

- Docker
- AWS / Azure

Libraries & Frameworks

- FastAPI
- React.js
- Next.js
- Tailwind CSS
- JWT Authentication

- Chart.js
- Plotly

Dev & Deployment Tools

- VS Code
 - Git & GitHub
 - Docker & Docker Compose
 - GitHub Actions
 - Postman
-

8. Performance Metrics

Funding Recommendation Metrics

- Recommendation precision
- Recommendation recall
- Eligibility matching accuracy

Research Intelligence Metrics

- Trend detection accuracy
- Emerging topic identification accuracy
- Research relevance score

Patent Intelligence Metrics

- Patent clustering quality
- Patent relevance accuracy
- Technology mapping effectiveness

Innovation Analytics Metrics

- Innovation scoring consistency

- Commercialization recommendation quality
- Technology opportunity identification rate

System Performance Metrics

- API response time
 - Dashboard loading speed
 - Search response latency
 - Concurrent user handling capacity
-

9. Example Quantitative Goals

Funding Discovery

Recommend highly relevant funding opportunities based on research interests and innovation domains.

Research Intelligence

Identify emerging research trends and innovation opportunities across multiple disciplines.

Patent Intelligence

Generate comprehensive patent landscape analysis and competitor intelligence.

Commercialization Recommendations

Provide actionable commercialization and innovation strategies for researchers and startups.

Platform Performance

Support large-scale research, patent, and funding analytics while maintaining stable performance and responsiveness.